Sean Shiroma

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Education

Stony Brook University — Stony Brook, NY
B.S. in Computer Science; B.S. in Applied Mathematics & Statistics — *May 2027*

- GPA 3.94/4.0; Honors program in Computer Science
- Coursework: Grad. Computer Architecture, Cloud Computing, Networks, Grad. Operating Systems, DSA

Experience

Software Engineer — Sep. 2026 – Present
Lukacsy-Bhowmik Trading — *Stony Brook, NY*

- Build an autonomous Novig Sports Market Making System in Python, achieving $400k volume per week
- Implement quoting, market discovery, order management, and reconciliation; coordinated through Redis

Technology Application Development Intern — June 2026 – Aug. 2026
Citi — *New York, NY*

- Developed the latency and throughput observability platform for the exchange-traded derivatives clearing system that processes 2 million trades/day
- Instrumented three Spring microservices with OpenTelemetry; streamed through the OTel collector into Kafka
- Computed trading latency and throughput metrics with Flink, exported to Prometheus with Grafana dashboards

Software Engineer Intern — Feb. 2026 – May 2026
SRC Inc. — *Syracuse, NY (Remote)*

- Developed a performance monitoring platform in Python and C++ that pinpoints faults across radar systems
- Implemented the collector feeding InfluxDB, designed the schema, and built fault classification and alerting

Research Fellow — Aug. 2025 – Present
File Systems & Storage Lab — *Stony Brook, NY*

- Engineered a distributed, decentralized key management system for century-scale cryptographic integrity in Python with MP-SPDZ for cryptography, gRPC for internode communication, and SQLite for storing shares
- Raised throughput from 1.9 to 10 req/s and reduced p95 latency from 13.6s to 2s by creating an auto-scaling worker pool for Beaver triple precomputation and modifying MP-SPDZ's C++ internals to support it

Software Engineer Intern — June 2025 – Aug. 2025
Broadridge Financial Solutions — *New York, NY*

- Implemented an Oracle disaster recovery database for a bank's financial data platform, porting Snowflake objects
- Configured the Oracle repository layer using MyBatis and HikariCP within Spring-based services

Projects

Polymarket Weather Trading System | *Python, PyTorch, NumPy, CUDA* — June 2026 – Present

- Develop a gradient-boosted model and a 24M-parameter transformer trained on >1 TB of minute-level observations from 900+ US weather stations, producing calibrated probability distributions for each market outcome
- Build an event-driven backtester on NautilusTrader with order book replay and a custom fill model for thin books
- Implement the order management system for pricing, order placement, reconciliation, and position tracking

RISC-V Operating System Kernel | *C, QEMU, GDB, RISC-V Toolchain* — Jan. 2026 – May 2026

- Built a single-core RISC-V OS in C modeled on Linux and FreeBSD with trap and interrupt handling, multi-level paging with lazy allocation and copy-on-write, a filesystem, preemptive round-robin scheduling; boots in QEMU
- Implemented 60+ Unix-like syscalls for process control, memory management, inter-process communication, and I/O

Technical Skills

Languages: Java, Go, C, C++, Python, JavaScript/TypeScript, SQL
Frameworks/Tools: Spring (Boot, Batch), gRPC, Apache Kafka, Apache Flink, OpenTelemetry, Django, Node.js
Cloud/DevOps: AWS, GCP, Docker, Kubernetes, Nginx, Jenkins, Grafana, Prometheus
Databases: Oracle, Snowflake, PostgreSQL, ScyllaDB, MongoDB, Redis, SQLite, InfluxDB

Awards

Competitions: Top 5 Jane Street GUTS++ NYC (2025), Metrobank Math Challenge Regional Finalist (2018–2019)
Scholarships: Weinig Foundation, Joan Kenny Endowed, General Engineering, Presidential